Title

Othar Zaldastani II

Date

---

1. Load & Reshape

The raw data is in wide format — one column per year. We pivot to long, recode series names, then pivot back to get one row per country-year with each indicator as its own column.

```
## Rows: 6,076
## Columns: 17
## $ country           <chr> "Afghanistan", "Afghanistan", "Afghanistan", "Afghan~
## $ iso3              <chr> "AFG", "AFG", "AFG", "AFG", "AFG", "AFG", "AFG", "AF~
## $ year              <int> 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004~
## $ nat_resources     <dbl> NA, NA, NA, NA, NA, NA, 1.2761488, 0.7313126, 0.4589~
## $ corruption        <dbl> -1.291705, NA, -1.176012, NA, -1.271724, NA, -1.2511~
## $ gov_effectiveness <dbl> -2.1751666, NA, -2.1022918, NA, -2.1739461, NA, -1.5~
## $ pol_stability     <dbl> -2.417310, NA, -2.427355, NA, -2.438969, NA, -2.0350~
## $ gini              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ tax_revenue       <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, 6.967598, 5.~
## $ gov_expense       <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, 20.57817, 24~
## $ rule_of_law       <dbl> -1.788075, NA, -1.734887, NA, -1.780661, NA, -1.6734~
## $ reg_quality       <dbl> -2.090330, NA, -2.062872, NA, -2.080253, NA, -1.8115~
## $ literacy          <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ energy_use        <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ trade             <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ gdp_pc            <dbl> NA, NA, NA, NA, 308.3183, 277.1181, 338.1400, 346.07~
## $ fdi               <dbl> NA, NA, NA, NA, 0.004827601, 0.024168568, 1.30694987~
```

---

2. Feature Engineering

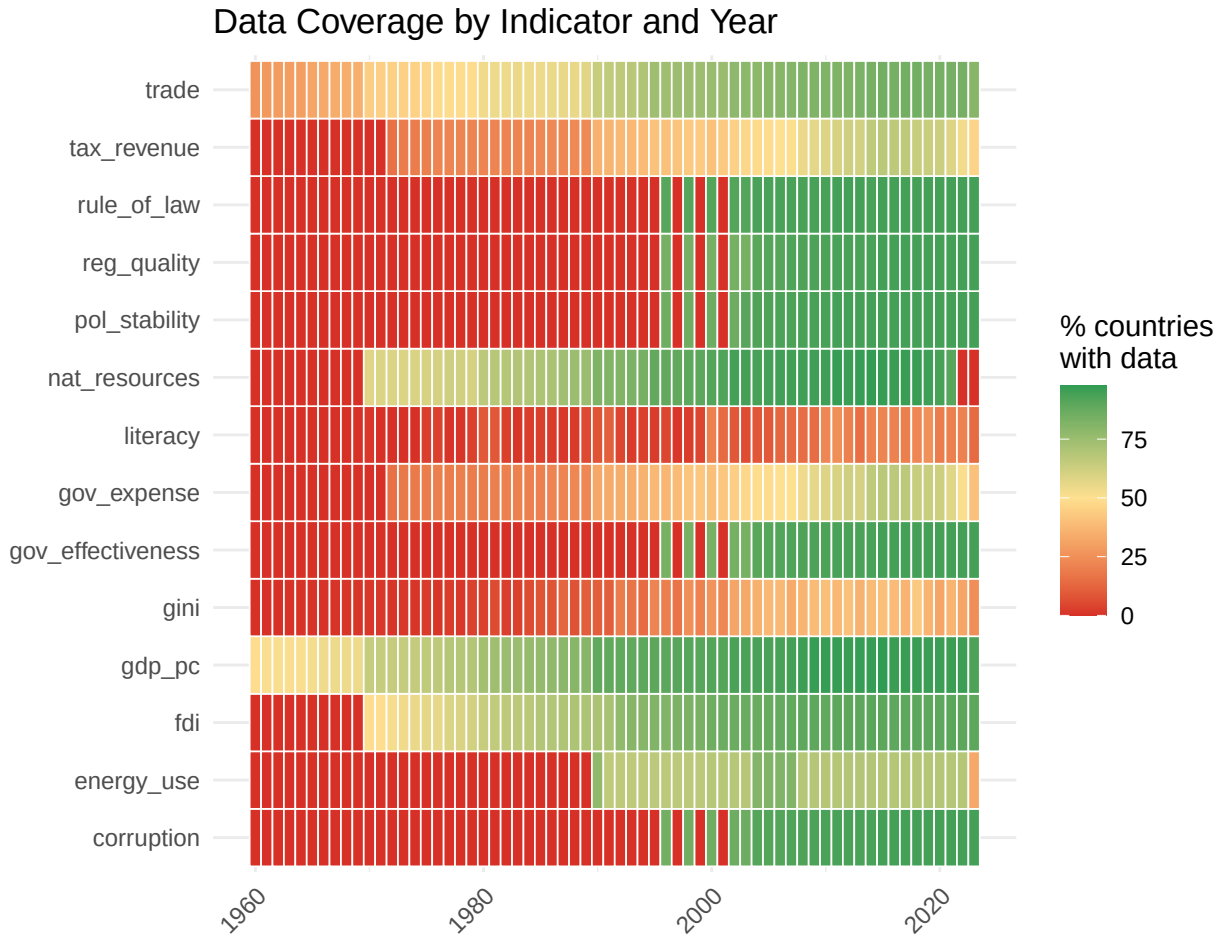
Feature	Description
`log_gdp_pc`	Log GDP per capita — stabilises variance
`gdp_lag1/3`	Lagged GDP — autoregressive predictor
`fdi_lag1/3`	Lagged FDI — investment effects are delayed
`gov_composite`	Mean of 5 World Bank governance indicators
`gov_roll3`	3-year rolling governance mean
`fdi_x_gov`	FDI x governance interaction
`trade_lag1`	Lagged trade openness

---

3. Exploratory Data Analysis

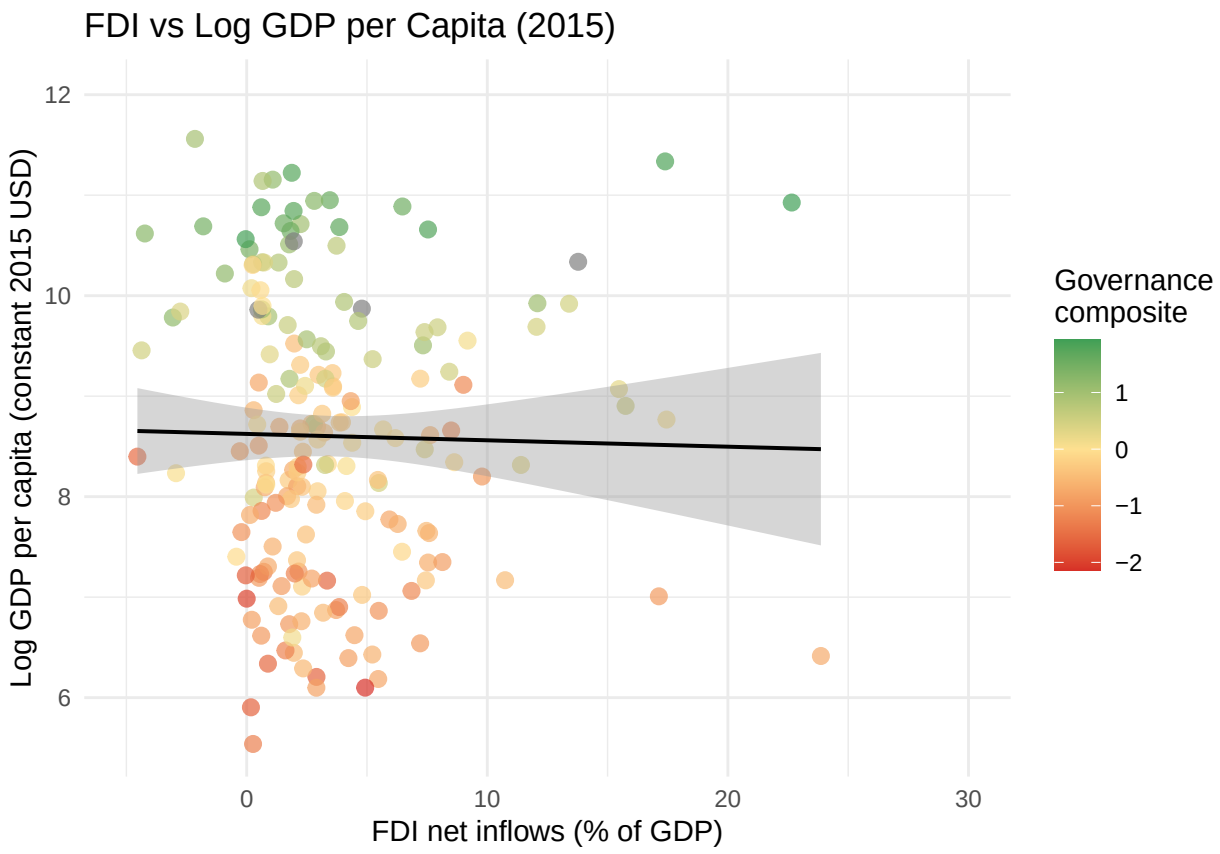
3.1 Data Coverage

Red = sparse, green = good coverage.

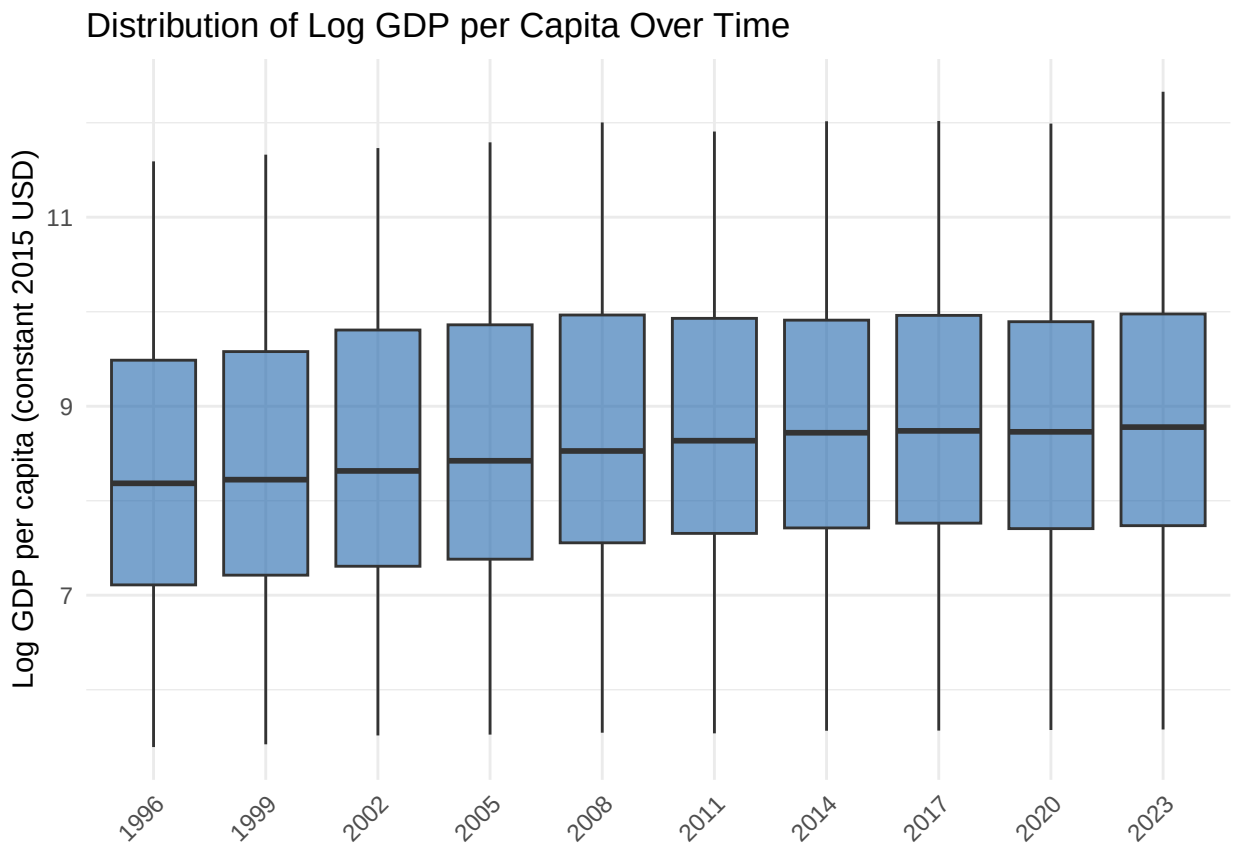


3.2 FDI vs Log GDP per Capita

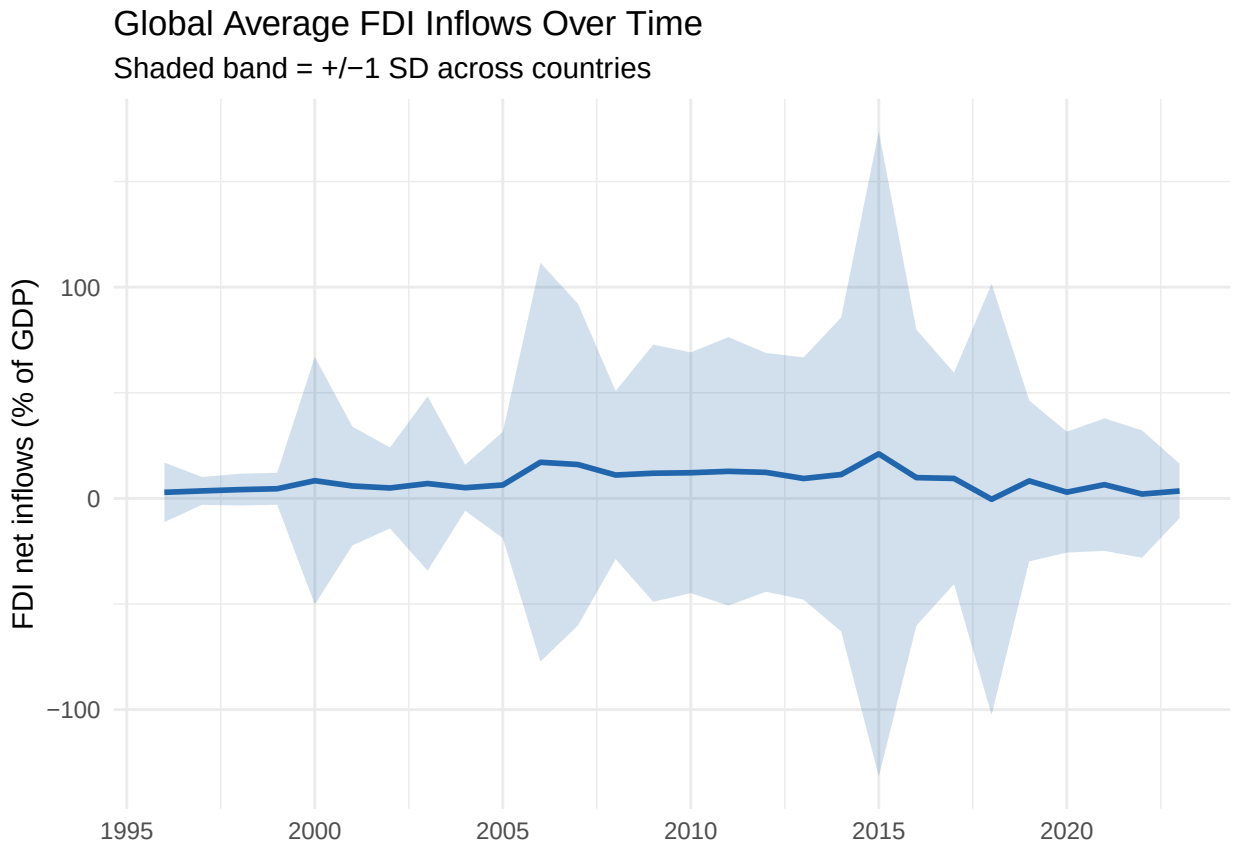
Colour shows governance quality — higher governance countries tend to attract more productive FDI.


FDI vs Log GDP per Capita (2015)
Log GDP per capita (constant 2015 USD)
12
10
8
6
0
10
20
30
FDI net inflows (% of GDP)
Governance composite
1
0
−1
−2

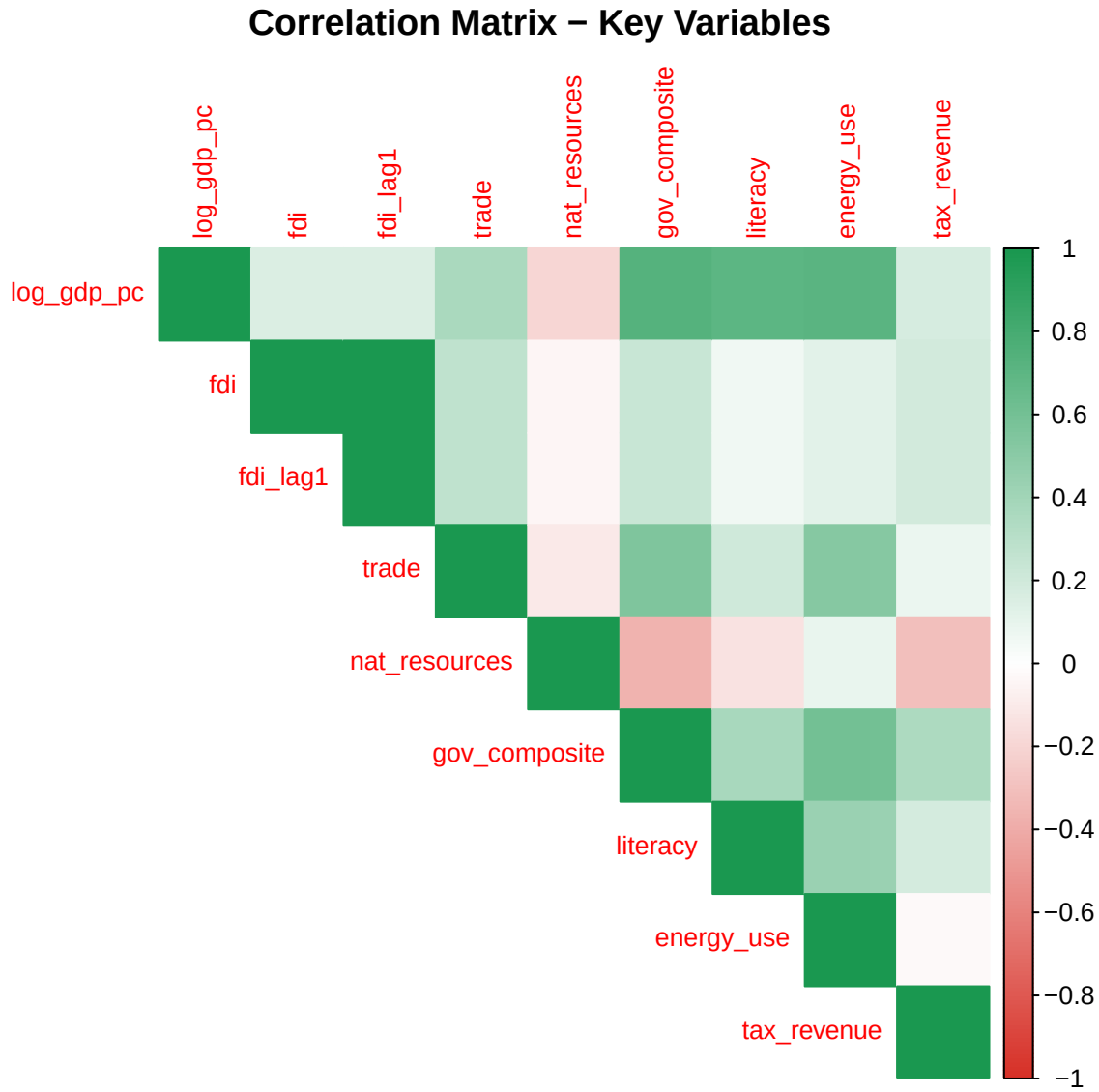
3.3 GDP per Capita Distribution Over Time



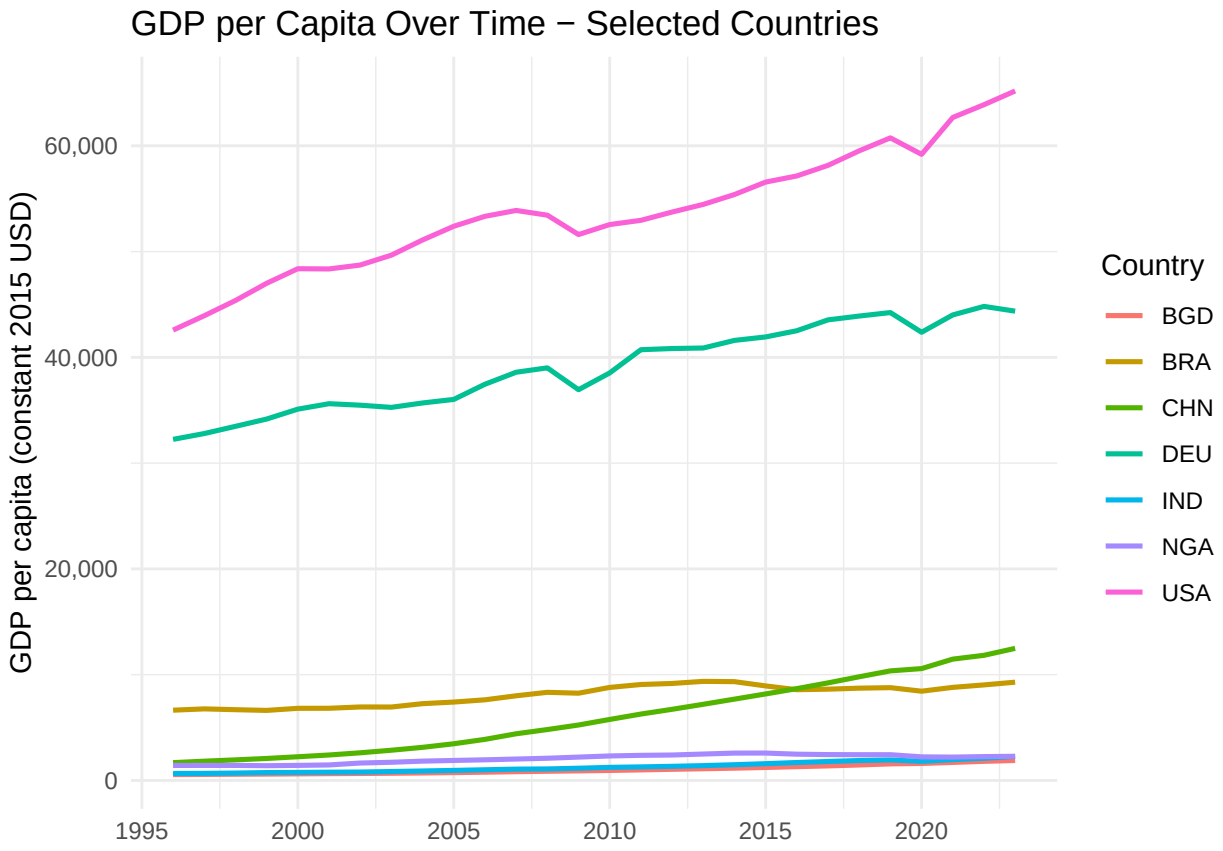
3.4 Global FDI Trend



3.5 Correlation Matrix



3.6 GDP Trends — Selected Countries



---

4. Panel Regression (Fixed Effects)

Two-way fixed effects control for time-invariant country differences and global year shocks. Standard errors are clustered by country.

Model 1 - FDI Only

```
##
## t test of coefficients:
##
##            Estimate Std. Error t value Pr(>|t|)
## fdi_lag1 0.00026175 0.00029965  0.8735    0.383
```

Model 2 - FDI + Macro Controls

```
##
## t test of coefficients:
##
##                  Estimate  Std. Error t value Pr(>|t|)
## fdi_lag1       0.00024040  0.00026795  0.8972 0.370269
## trade         -0.00018979  0.00084356 -0.2250 0.822128
## nat_resources  0.00082687  0.00176165  0.4694 0.639116
## tax_revenue    0.01106695  0.00649162  1.7048 0.089172 .
```

```
## gov_expense   -0.00959274  0.00347148 -2.7633 0.006043 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Model 3 - Full Specification

```
##
## t test of coefficients:
##
##                  Estimate  Std. Error t value  Pr(>|t|)
## fdi_lag1       0.00383522  0.00226442  1.6937 0.0912762 .
## fdi_x_gov     -0.00352131  0.00206914 -1.7018 0.0897392 .
## gov_composite  0.26390181  0.04664513  5.6577 3.354e-08 ***
## trade          0.00048160  0.00066470  0.7245 0.4692552
## nat_resources -0.00047302  0.00180283 -0.2624 0.7931967
## tax_revenue    0.00128770  0.00365953  0.3519 0.7251577
## energy_use     0.00021082  0.00006123  3.4431 0.0006499 ***
## literacy       0.00613710  0.00250110  2.4538 0.0146584 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Model Fit Summary

Model	Within R2
1: FDI only	0.0009
2: + Macro	0.0610
3: Full	0.3991

---

5. Machine Learning - XGBoost

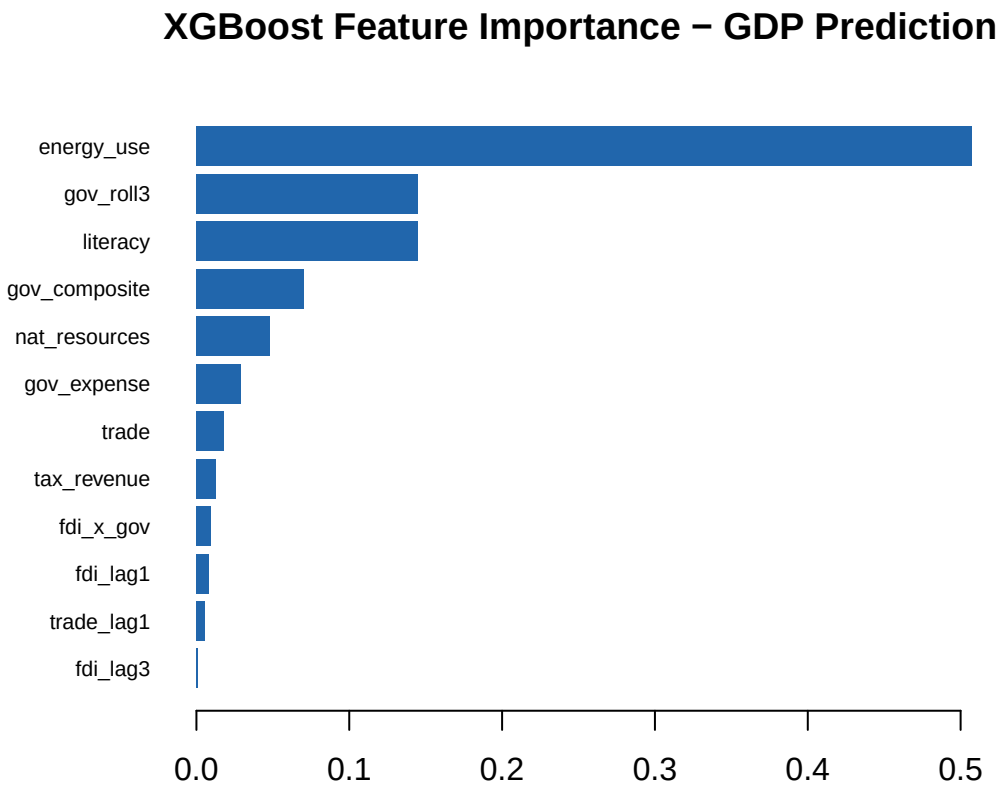
Time-aware split to prevent data leakage: train on 1960-2015, validate on 2016-2019, test on 2020+.

```
## [1]  train-rmse:7.625030 valid-rmse:7.639737
## Multiple eval metrics are present. Will use valid_rmse for early stopping.
## Will train until valid_rmse hasn't improved in 30 rounds.
##
## [101]    train-rmse:0.147456 valid-rmse:0.356224
## [201]    train-rmse:0.068504 valid-rmse:0.326062
## [301]    train-rmse:0.041356 valid-rmse:0.323676
## [401]    train-rmse:0.026568 valid-rmse:0.321344
## Stopping. Best iteration:
## [447]    train-rmse:0.022218 valid-rmse:0.320831
```

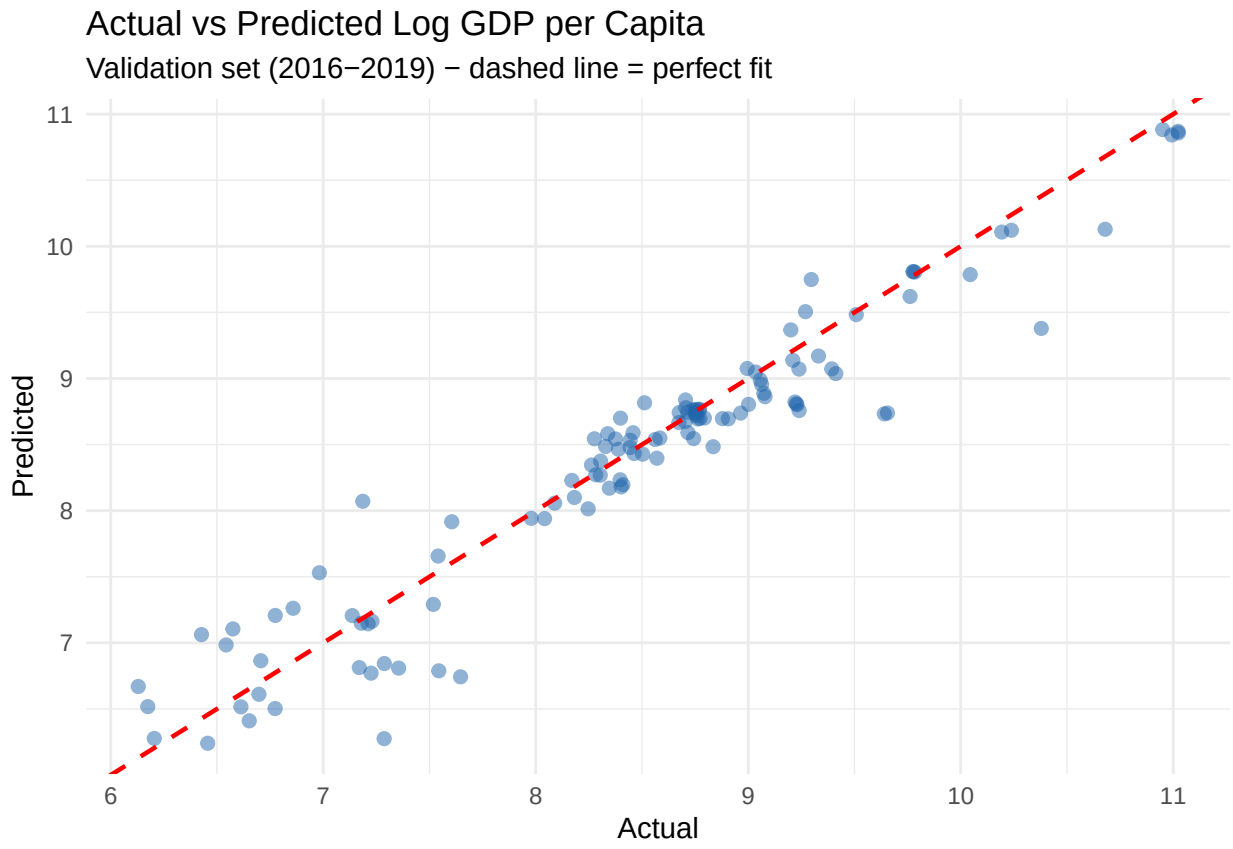
Performance

Split	RMSE	MAE
Validation (2016-2019)	0.3208	0.2227
Test (2020+)	0.3985	0.3027

Feature Importance



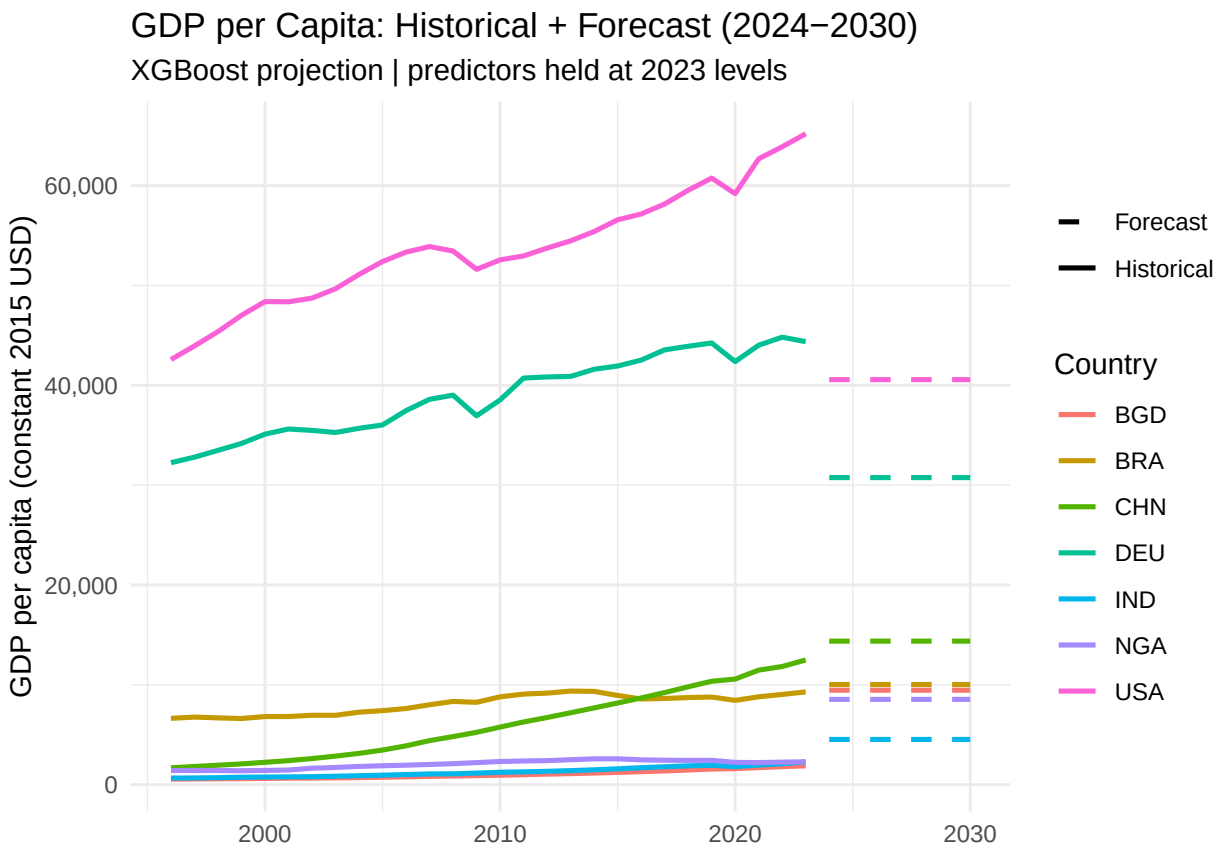
Actual vs Predicted



---

6. Forecast 2024-2030

Each country's predictors held at their 2023 values. Missing values imputed with the global median.



Top 10 Countries by Projected GDP per Capita in 2030

Country	Projected GDP pc (2030)
Singapore	$52,817
United States	$40,563
Korea, Rep.	$37,208
Australia	$36,696
Finland	$36,141
Canada	$35,423
Iceland	$35,136
Norway	$34,988
Luxembourg	$33,091
Germany	$30,752

---

7. Export Results

Code Appendix

```r
# Setup Chunk
knitr::opts_chunk$set(
  echo = FALSE,
  warning = FALSE,
  message = FALSE,
  fig.align = "center",
  attr.source = '.numberLines',
  cache = FALSE)

library(tidyverse)
library(psych)
library(psychTools)
library(dplyr)
library(knitr)
library(kableExtra)
library(epitools)
library(broom)
library(modelsummary)
library(kableExtra)
set.seed(24)

options(
  modelsummary_stars = TRUE,
  modelsummary_output = "markdown",
  modelsummary_shape = model ~ term,
  modelsummary_fmt = 2,
  modelsummary_gof_omit = "")

ms <- function(models, ...) {
  g <- broom::glance(models)
  modelsummary(models, stars = F, output = "markdown",
               shape = model ~ term, fmt = 2,
               statistic = c("({std.error})", "p = {p.value}"),
               notes = paste0("R² = ", round(g$r.squared, 3),
                              ", Adj. R² = ", round(g$adj.r.squared, 3),
                              ", F = ", round(g$statistic, 2),
                              ", p = ", format.pval(g$p.value, digits = 3),
                              ", N = ", g$nobs), ...)}

kable2 <- function(x, ...) {
  kableExtra::kable_styling(
    knitr::kable(x, ...))}


library(tidyverse)
library(plm)
library(lmtest)
library(sandwich)
library(xgboost)
library(Metrics)
library(corrplot)
library(scales)
```

```r
library(zoo)
library(knitr)
library(kableExtra)
FDI_Data4 <- read.csv("~/Downloads/P_Data_Extract_From_World_Development_Indicators-2/e24a155d-2435-4e2
raw <- FDI_Data4
names(raw) <- gsub("\\.", " ", names(raw)) |> trimws()

long <- raw %>%
  pivot_longer(
    cols      = matches("^X\\d{4}"),
    names_to  = "year_label",
    values_to = "value"
  ) %>%
  mutate(
    year  = as.integer(str_extract(year_label, "\\d{4}")),
    value = suppressWarnings(as.numeric(value)),
    var   = case_when(
      `Series Name` == "Total natural resources rents (% of GDP)"                 ~ "nat_resources",
      `Series Name` == "Control of Corruption: Estimate"                          ~ "corruption",
      `Series Name` == "Government Effectiveness: Estimate"                       ~ "gov_effectiveness
      `Series Name` == "Political Stability and Absence of Violence/Terrorism: Estimate" ~ "pol_stabili
      `Series Name` == "Gini index"                                               ~ "gini",
      `Series Name` == "Tax revenue (% of GDP)"                                   ~ "tax_revenue",
      `Series Name` == "Expense (% of GDP)"                                       ~ "gov_expense",
      `Series Name` == "Rule of Law: Estimate"                                    ~ "rule_of_law",
      `Series Name` == "Regulatory Quality: Estimate"                             ~ "reg_quality",
      `Series Name` == "Literacy rate, adult total (% of people ages 15 and above)" ~ "literacy",
      grepl("Human capital", `Series Name`)                                       ~ "hci",
      `Series Name` == "Energy use (kg of oil equivalent per capita)"             ~ "energy_use",
      `Series Name` == "Trade (% of GDP)"                                         ~ "trade",
      `Series Name` == "GDP per capita (constant 2015 US$)"                       ~ "gdp_pc",
      `Series Name` == "Foreign direct investment, net inflows (% of GDP)"        ~ "fdi",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(!is.na(var)) %>%
  select(`Country Name`, `Country Code`, year, var, value)

panel <- long %>%
  pivot_wider(
    id_cols     = c(`Country Name`, `Country Code`, year),
    names_from  = var,
    values_from = value,
    values_fn   = mean
  ) %>%
  rename(country = `Country Name`, iso3 = `Country Code`) %>%
  rename_with(~ gsub("[^a-zA-Z0-9_]", "_", .x)) %>%
  filter(year >= 1996, year <= 2023)

glimpse(panel)
panel <- panel %>%
  arrange(iso3, year) %>%
  group_by(iso3) %>%
```

```r
  mutate(
    # Log GDP - stabilises variance for modelling
    log_gdp_pc    = log(gdp_pc),

    # GDP growth rate
    gdp_growth    = (gdp_pc - lag(gdp_pc)) / lag(gdp_pc) * 100,

    # Lagged GDP - autoregressive signal
    gdp_lag1      = lag(gdp_pc, 1),
    gdp_lag3      = lag(gdp_pc, 3),

    # Lagged FDI - investment effects on GDP are delayed
    fdi_lag1      = lag(fdi, 1),
    fdi_lag3      = lag(fdi, 3),

    # Governance composite - mean of 5 WB indicators
    gov_composite = rowMeans(cbind(corruption, gov_effectiveness,
                                   pol_stability, rule_of_law,
                                   reg_quality), na.rm = TRUE),

    # Rolling 3-year governance - smooths year-to-year noise
    gov_roll3     = rollmeanr(gov_composite, 3, fill = NA),

    # FDI x governance - FDI matters more in high-governance environments
    fdi_x_gov     = fdi_lag1 * gov_composite,

    # Trade
    trade_lag1    = lag(trade, 1),
    trade_growth  = (trade - lag(trade)) / lag(trade) * 100,

    # Resource dependency lag
    nat_res_lag1  = lag(nat_resources, 1),

    # Tax burden change
    tax_change    = tax_revenue - lag(tax_revenue)
  ) %>%
  ungroup()
coverage <- long %>%
  filter(year >= 1960, year <= 2023) %>%
  group_by(var, year) %>%
  summarise(pct_available = mean(!is.na(value)) * 100, .groups = "drop")

ggplot(coverage, aes(x = year, y = var, fill = pct_available)) +
  geom_tile(colour = "white", linewidth = 0.3) +
  scale_fill_gradient2(
    low      = "#d73027", mid = "#fee090", high = "#1a9850",
    midpoint = 50, name = "% countries\nwith data"
  ) +
  labs(title = "Data Coverage by Indicator and Year", x = NULL, y = NULL) +
  theme_minimal(base_size = 11) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
panel %>%
  filter(year == 2015, !is.na(fdi), !is.na(gdp_pc)) %>%
```

```r
  ggplot(aes(x = fdi, y = log_gdp_pc, colour = gov_composite)) +
  geom_point(alpha = 0.7, size = 2.5) +
  geom_smooth(method = "lm", se = TRUE, colour = "black", linewidth = 0.7) +
  scale_colour_gradient2(
    low      = "#d73027", mid = "#fee090", high = "#1a9850",
    midpoint = 0, name = "Governance\ncomposite"
  ) +
  scale_x_continuous(limits = c(-5, 30)) +
  labs(
    title = "FDI vs Log GDP per Capita (2015)",
    x     = "FDI net inflows (% of GDP)",
    y     = "Log GDP per capita (constant 2015 USD)"
  ) +
  theme_minimal()
panel %>%
  filter(!is.na(gdp_pc), year %in% seq(1960, 2023, by = 3)) %>%
  mutate(year = factor(year)) %>%
  ggplot(aes(x = year, y = log_gdp_pc)) +
  geom_boxplot(fill = "#2166ac", alpha = 0.6, outlier.size = 0.8) +
  labs(
    title = "Distribution of Log GDP per Capita Over Time",
    x     = NULL,
    y     = "Log GDP per capita (constant 2015 USD)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
panel %>%
  group_by(year) %>%
  summarise(mean_fdi = mean(fdi, na.rm = TRUE),
            sd_fdi   = sd(fdi, na.rm = TRUE)) %>%
  ggplot(aes(x = year, y = mean_fdi)) +
  geom_ribbon(aes(ymin = mean_fdi - sd_fdi, ymax = mean_fdi + sd_fdi),
              alpha = 0.2, fill = "#2166ac") +
  geom_line(colour = "#2166ac", linewidth = 1) +
  labs(
    title    = "Global Average FDI Inflows Over Time",
    subtitle = "Shaded band = +/-1 SD across countries",
    x        = NULL,
    y        = "FDI net inflows (% of GDP)"
  ) +
  theme_minimal()
cor_vars <- panel %>%
  select(log_gdp_pc, fdi, fdi_lag1, trade, nat_resources,
         gov_composite, literacy, energy_use, tax_revenue) %>%
  drop_na()

corrplot(
  cor(cor_vars),
  method = "color",
  type   = "upper",
  tl.cex = 0.85,
  col    = colorRampPalette(c("#d73027", "white", "#1a9850"))(200),
  title  = "Correlation Matrix - Key Variables",
```

```r
  mar    = c(0, 0, 2, 0)
)
showcase <- c("USA", "CHN", "IND", "BRA", "DEU", "NGA", "BGD")

panel %>%
  filter(iso3 %in% showcase, !is.na(gdp_pc)) %>%
  ggplot(aes(x = year, y = gdp_pc, colour = iso3)) +
  geom_line(linewidth = 0.9) +
  scale_y_continuous(labels = comma) +
  labs(
    title  = "GDP per Capita Over Time - Selected Countries",
    x      = NULL,
    y      = "GDP per capita (constant 2015 USD)",
    colour = "Country"
  ) +
  theme_minimal()
model_data <- panel %>%
  select(country, iso3, year, log_gdp_pc, fdi_lag1, fdi_lag3,
         fdi_x_gov, trade, nat_resources, gov_composite,
         trade_lag1, literacy, energy_use, tax_revenue, gov_expense) %>%
  drop_na() %>%
  filter(is.finite(log_gdp_pc))

pdata <- pdata.frame(model_data, index = c("iso3", "year"))

# Model 1: FDI only
fe1 <- plm(log_gdp_pc ~ fdi_lag1,
           data = pdata, model = "within", effect = "twoways")

# Model 2: FDI + macro controls
fe2 <- plm(log_gdp_pc ~ fdi_lag1 + trade + nat_resources +
             tax_revenue + gov_expense,
           data = pdata, model = "within", effect = "twoways")

# Model 3: Full specification
fe3 <- plm(log_gdp_pc ~ fdi_lag1 + fdi_lag3 + fdi_x_gov +
             gov_composite + trade + trade_lag1 + nat_resources +
             tax_revenue + energy_use + literacy,
           data = pdata, model = "within", effect = "twoways")

robust_se <- function(model) {
  coeftest(model, vcov = vcovHC(model, type = "HC1", cluster = "group"))
}
robust_se(fe1)
robust_se(fe2)
robust_se(fe3)
tibble(
  Model       = c("1: FDI only", "2: + Macro", "3: Full"),
  `Within R2` = c(
    round(summary(fe1)$r.squared["rsq"], 4),
    round(summary(fe2)$r.squared["rsq"], 4),
    round(summary(fe3)$r.squared["rsq"], 4)
  )
```

```r
) %>%
  kable() %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
features <- c("fdi_lag1", "fdi_lag3", "fdi_x_gov",
              "trade", "trade_lag1", "nat_resources", "gov_composite",
              "gov_roll3", "literacy", "energy_use",
              "tax_revenue", "gov_expense")

ml_data <- panel %>%
  select(year, log_gdp_pc, all_of(features)) %>%
  drop_na() %>%
  filter(is.finite(log_gdp_pc))

train <- ml_data %>% filter(year <= 2015)
valid <- ml_data %>% filter(year > 2015, year <= 2019)
test  <- ml_data %>% filter(year >= 2020)

make_matrix <- function(df) as.matrix(df %>% select(all_of(features)))

dtrain <- xgb.DMatrix(make_matrix(train), label = train$log_gdp_pc)
dvalid <- xgb.DMatrix(make_matrix(valid), label = valid$log_gdp_pc)
dtest  <- xgb.DMatrix(make_matrix(test),  label = test$log_gdp_pc)
params <- list(
  objective        = "reg:squarederror",
  eta              = 0.05,
  max_depth        = 4,
  subsample        = 0.8,
  colsample_bytree = 0.8,
  min_child_weight = 5
)

set.seed(42)
xgb_model <- xgb.train(
  params                = params,
  data                  = dtrain,
  nrounds               = 500,
  watchlist             = list(train = dtrain, valid = dvalid),
  early_stopping_rounds = 30,
  print_every_n         = 100,
  verbose               = 1
)
pred_valid <- predict(xgb_model, dvalid)
pred_test  <- predict(xgb_model, dtest)

tibble(
  Split = c("Validation (2016-2019)", "Test (2020+)"),
  RMSE  = c(round(rmse(valid$log_gdp_pc, pred_valid), 4),
            round(rmse(test$log_gdp_pc,  pred_test),  4)),
  MAE   = c(round(mae(valid$log_gdp_pc,  pred_valid), 4),
            round(mae(test$log_gdp_pc,   pred_test),  4))
) %>%
  kable() %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

```r
importance <- xgb.importance(feature_names = features, model = xgb_model)
xgb.plot.importance(importance,
                    main = "XGBoost Feature Importance - GDP Prediction",
                    col  = "#2166ac")
tibble(
  actual    = valid$log_gdp_pc,
  predicted = pred_valid
) %>%
  ggplot(aes(x = actual, y = predicted)) +
  geom_point(alpha = 0.5, colour = "#2166ac", size = 2) +
  geom_abline(slope = 1, intercept = 0, colour = "red",
              linetype = "dashed", linewidth = 0.8) +
  labs(
    title    = "Actual vs Predicted Log GDP per Capita",
    subtitle = "Validation set (2016-2019) - dashed line = perfect fit",
    x        = "Actual",
    y        = "Predicted"
  ) +
  theme_minimal()
latest <- panel %>%
  filter(year == 2023) %>%
  select(iso3, country, all_of(features))

for (v in features) {
  med <- median(latest[[v]], na.rm = TRUE)
  latest[[v]][is.na(latest[[v]])] <- med
}

forecast_rows <- expand_grid(iso3 = latest$iso3, year = 2024:2030) %>%
  left_join(latest, by = "iso3")

dforecast <- xgb.DMatrix(as.matrix(forecast_rows %>% select(all_of(features))))
forecast_rows$pred_log_gdp <- predict(xgb_model, dforecast)
forecast_rows$pred_gdp_pc  <- exp(forecast_rows$pred_log_gdp)
historical <- panel %>%
  filter(iso3 %in% showcase, !is.na(gdp_pc)) %>%
  select(iso3, year, gdp_pc) %>%
  mutate(type = "Historical")

projected <- forecast_rows %>%
  filter(iso3 %in% showcase) %>%
  select(iso3, year, gdp_pc = pred_gdp_pc) %>%
  mutate(type = "Forecast")

bind_rows(historical, projected) %>%
  ggplot(aes(x = year, y = gdp_pc, colour = iso3,
             linetype = type, group = interaction(iso3, type))) +
  geom_line(linewidth = 0.9) +
  scale_y_continuous(labels = comma) +
  scale_linetype_manual(values = c("Historical" = "solid", "Forecast" = "dashed")) +
  labs(
    title    = "GDP per Capita: Historical + Forecast (2024-2030)",
    subtitle = "XGBoost projection | predictors held at 2023 levels",
```

```
    x        = NULL,
    y        = "GDP per capita (constant 2015 USD)",
    colour   = "Country",
    linetype = NULL
  ) +
  theme_minimal()
forecast_rows %>%
  filter(year == 2030) %>%
  arrange(desc(pred_gdp_pc)) %>%
  select(Country = country, `Projected GDP pc (2030)` = pred_gdp_pc) %>%
  head(10) %>%
  mutate(`Projected GDP pc (2030)` = dollar(round(`Projected GDP pc (2030)`))) %>%
  kable() %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
write.csv(panel,         "FDI_GDP_panel_clean.csv",        row.names = FALSE)
write.csv(forecast_rows, "FDI_GDP_forecast_2024_2030.csv", row.names = FALSE)

cat("Saved: FDI_GDP_panel_clean.csv and FDI_GDP_forecast_2024_2030.csv")
```

Attributions: I have used some template/styling code from GitHub user alexpghayes.